

Brownian Motion & Itô Calculus

Session 19 • The foundation everyone needs — intuition for both tracks

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

Graduate Finance Course • Spring 2027 • Session 19 of 32

What we'll cover today

1

Random walks → Brownian motion

From discrete to continuous

2

Properties of Brownian motion

Continuous but nowhere differentiable

3

Why ordinary calculus fails

The $dW^2 = dt$ trick

4

Itô's lemma

Stochastic chain rule

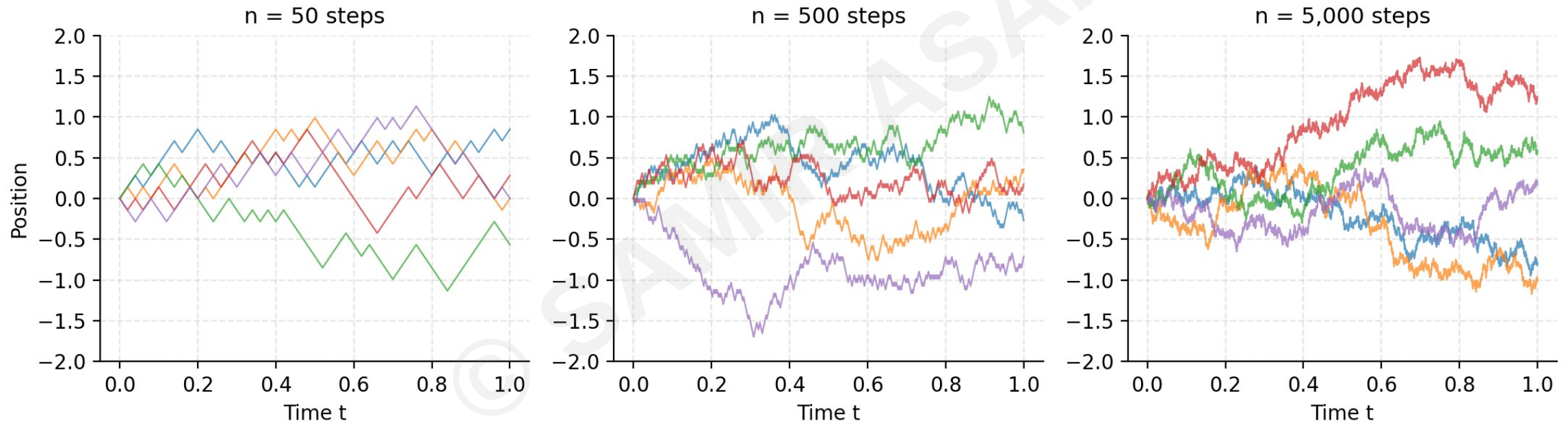
5

Application: discount factor dynamics

Where we'll use this in GE-LAV

From random walks to Brownian motion

Random walks → Brownian motion as step size shrinks



Brownian motion $W(t)$ is the limit

Three defining properties:

- $W(0) = 0$
- $W(t) - W(s) \sim N(0, t-s)$
- Independent increments

Why ordinary calculus fails on Brownian motion

Brownian motion is continuous everywhere but differentiable nowhere. Standard calculus doesn't apply.

The fundamental trick: $(dW)^2 = dt$, not 0

In ordinary calculus, $(dt)^2 \rightarrow 0$ faster than $dt \rightarrow 0$, so we ignore it. In stochastic calculus, the variance of dW grows like $\sqrt{dt}$, so $(dW)^2 \sim dt$ — same order!

STANDARD CHAIN RULE

$$df = f'(x) dx$$

Linear in dx (one term)

ITÔ'S LEMMA

$$df = f'(x) dx + \frac{1}{2} f''(x) (dx)^2$$

Extra second-order term (the Itô correction)

Application: where Itô shows up in GE-LAV

USE 1

OU process derivation

The OU SDE $dL = \kappa(\bar{L} - L)dt + \sigma dW$ is solved using Itô's lemma applied to the integrating factor $\exp(\kappa t) \cdot L(t)$.

→ Session 25 (Track 2)

USE 2

Discount factor dynamics

Applying Itô to $f(r) = \exp(-rT)$ reveals the Jensen convexity correction term — the source of Failure 2.

→ Session 24 (Jensen)

USE 3

HJB equation derivation

The Hamilton-Jacobi-Bellman equation for optimal exit timing comes from applying Itô to the value function $V(L, t)$.

→ Session 20 (HJB)

Track 2 students: read Karatzas-Shreve Ch.3-4. Track 1 students: intuition above is sufficient.

Session 19 summary

What we accomplished today

- 1 Brownian motion is the continuous limit of random walks — fundamental object of stochastic calculus
- 2 BM is continuous but nowhere differentiable. Standard calculus doesn't apply.
- 3 Itô's lemma is the stochastic chain rule — adds a second-order term ($\frac{1}{2}f'' \cdot (dx)^2$)
- 4 GE-LAV uses Itô in three places: OU derivation, Jensen correction, HJB equation

Next session

Session 20: HJB equation and optimal stopping — the math behind $L^*(t)$